LAB 7
Configuration of Switch, VLAN configuration and Inter-VLAN Routing

OBJECTIVES

- To design and configure multiple VLANs on a network switch.
- To set up trunk links and correctly assign switch ports to their respective VLANs.
- To implement inter-VLAN routing through a router and verify its operation.
- To perform connectivity tests to ensure devices in different VLANs can communicate properly.

THEORY

1. Switch Configuration

A switch is a Layer 2 networking device used to connect multiple devices within a local area network. It forwards frames based on MAC addresses, enabling efficient and targeted communication between connected hosts.

1.1 Switch Setup

Switch configuration generally includes tasks such as:

- Configuring and managing interfaces
- Enabling or disabling switch ports
- Assigning ports to appropriate VLANs

1.2 Significance of Proper Switch Configuration

A properly configured switch:

- Minimizes network collisions
- Enhances overall performance
- Helps organize and control network traffic efficiently

2. Virtual Local Area Network (VLAN)

A VLAN is a logical method of dividing a physical network into separate broadcast domains. Devices can be grouped based on role, department, or function regardless of their physical placement. VLANs improve efficiency and strengthen security by isolating network segments.

2.1 VLAN Setup

Typical VLAN configuration involves:

- Creating VLAN IDs on the switch
- Assigning ports to specific VLANs
- Configuring trunk links to transport multiple VLANs
- Enabling routing between VLANs

2.2 Importance of VLANs

Implementing VLANs helps to:

- Reduce unnecessary broadcast traffic
- Improve scalability and performance
- Increase security through network segmentation
- Simplify network management

3. Inter-VLAN Routing

Inter-VLAN Routing allows devices located in different VLANs to communicate with each other. Since VLANs operate as separate broadcast domains, routing is required to pass traffic between them. This can be achieved using a router or a Layer 3 switch.

3.1 Inter-VLAN Routing Techniques

- **Router-on-a-Stick (ROAS):** Uses one physical router interface with multiple sub-interfaces mapped to VLANs via trunking.
- **Switch Virtual Interface (SVI):** Creates logical Layer 3 interfaces on a multilayer switch for direct routing.
- **Routed Ports:** Configures physical switch ports as Layer 3 interfaces for point-to-point routing.

3.2 Importance of Inter-VLAN Routing

Proper implementation of inter-VLAN routing:

- Enables communication between different departments or network groups
- Maintains logical separation while allowing controlled data exchange
- Improves flexibility and scalability of the network

NETWORK TOPOLOGY

The topology represents a basic LAN environment where several PCs are connected to a switch, and the switch is linked to a router to enable communication beyond the local network. The switch is responsible for managing internal connectivity among the devices, while the router provides routing capabilities and supports inter-VLAN communication.

This type of setup is commonly used in laboratory or small network scenarios to demonstrate VLAN implementation and inter-VLAN routing, as it clearly shows how local switching and routing work together to enable efficient and controlled data exchange.

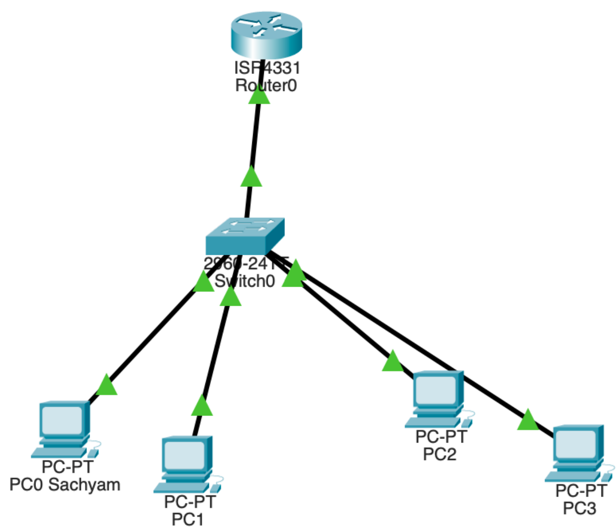

Fig: Network configuration

Testing

Ping Test: From PC1 to PC2

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.10.3

Pinging 192.168.10.3 with 32 bytes of data:

Request timed out.
Reply from 192.168.10.3: bytes=32 time<1ms TTL=127
Reply from 192.168.10.3: bytes=32 time<1ms TTL=127
Reply from 192.168.10.3: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.10.3:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Fig: Ping test

Configuration

VLAN and IP Addressing Scheme :

VLAN	Name	Network	Gateway
10	Computer	192.168.10.0/24	192.168.10.1
20	Electronics	192.168.20.0/24	192.168.20.1

Device IP configuration

Device	IPv4 Address	Subnet Mask	Default Gateway
PC0 Sachyam	192.168.10.2	255.255.255.0	192.168.10.1
PC1	192.168.10.3	255.255.255.0	192.168.10.1
PC2	192.168.20.2	255.255.255.0	192.168.20.1
PC3	192.168.20.3	255.255.255.0	192.168.20.1

COMMANDS

1. Switch Configuration

Create VLANs

```
Switch(config)# vlan 10
Switch(config-vlan)# name Computer
Switch(config-vlan)# exit

Switch(config)# vlan 20
Switch(config-vlan)# name Electronics
Switch(config-vlan)# exit
```

Assign Ports to VLANs

```
Switch(config)# interface range fa0/1-2
Switch(config-if-range)# switchport mode access
Switch(config-if-range)# switchport access vlan 10
Switch(config-if-range)# exit

Switch(config)# interface range fa0/3-4
Switch(config-if-range)# switchport mode access
Switch(config-if-range)# switchport access vlan 20
Switch(config-if-range)# exit
```

Configure Trunk Port (Switch <-> Router)

```
Switch(config)# interface fa0/5
Switch(config-if)# switchport mode trunk
Switch(config-if)# exit
```

Verification

```
Switch# show vlan brief
```

```
Switch>enable
Switch#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/6, Fa0/7, Fa0/8, Fa0/9
                                                Fa0/10, Fa0/11, Fa0/12, Fa0/13
                                                Fa0/14, Fa0/15, Fa0/16, Fa0/17
                                                Fa0/18, Fa0/19, Fa0/20, Fa0/21
                                                Fa0/22, Fa0/23, Fa0/24, Gig0/1
                                                Gig0/2
10   computer                         active    Fa0/1, Fa0/2
20   electronics                      active    Fa0/3, Fa0/4
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
Switch#
```

2. Router Configuration (Inter-VLAN Routing)

Sub-interface for VLAN 10

```
Router(config)# interface fa0/0.10
Router(config-subif)# encapsulation dot1Q 10
Router(config-subif)# ip address 192.168.10.1 255.255.255.0
Router(config-subif)# exit
```

Sub-interface for VLAN 10

```
Router(config)# interface fa0/0.20
Router(config-subif)# encapsulation dot1Q 20
Router(config-subif)# ip address 192.168.20.1 255.255.255.0
Router(config-subif)# exit
```

DISCUSSION

In this lab, a simple network topology consisting of a router, a switch, and multiple PCs was implemented. The switch provided internal connectivity among the devices, while the router enabled routing between VLANs and supported communication beyond the local network. By configuring VLANs, trunk links, and inter-VLAN routing, the experiment illustrated how network segmentation helps manage traffic efficiently and how routing enables controlled communication between separate VLANs.

CONCLUSION

This experiment successfully demonstrated the process of creating VLANs and configuring inter-VLAN routing in a network environment. The activity highlighted the importance of correct switch and router configuration in minimizing collisions, improving performance, and maintaining organized traffic flow. Overall, the lab provided a clear practical understanding of how VLANs and routing mechanisms work together to build efficient and scalable networks.